

D035

Educational Inquiry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A fifth-grade teacher at a school with a new outdoor classroom notices many students struggle with multi-step word problems. She wants to try a strategy she heard about at lunch. She has taught for nine years. According to the action research cycle, what should she do before trying the strategy?

- A. Wait until she has more experience.
- B. Try the strategy right away and see what happens.
- C. Define the problem with data and review scholarly literature on word-problem strategies before planning.
- D. Move word-problem lessons to the outdoor classroom.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A high school teacher who also coaches track drafts action research questions. The school recently won a regional award. Which question is most researchable in her own setting?

- A. "How did our school win the regional award?"
- B. "How can American education be improved?"
- C. "How does adding weekly peer review affect the quality of my tenth graders' argumentative essays over one semester?"
- D. "Why do some families not value education?"

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

For her literature review, a teacher who reads widely finds four sources on vocabulary instruction. One is on a popular teaching website with over a million followers. Which source provides the strongest research evidence?

- A. The popular teaching website.
- B. A peer-reviewed journal article reporting a controlled study.
- C. An opinion column by an experienced principal.
- D. A vendor's brochure citing its own results.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A teacher has cited only teacher blogs and a vendor white paper. What should she add to strengthen her literature review?

- A.** Peer-reviewed empirical studies on the same practice.
- B.** Nothing; all published sources are equal.
- C.** A second vendor white paper to balance the first.
- D.** More teacher blogs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A middle school counselor with a new office asks: “How do students who recently moved to our school describe their experience of making friends?” The school has 850 students. Which method fits best?

- A.** Counting all 850 students’ absences.
- B.** Comparing new students’ test scores with others’.
- C.** An experiment with random assignment to friendships.
- D.** Qualitative interviews or focus groups with new students.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A teacher asks: “By how many points did students’ quiz averages change after I added retrieval practice?” Which method fits?

- A.** A literature review only.
- B.** Quantitative comparison of quiz scores before and after.
- C.** Student interviews only.
- D.** A classroom observation journal only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A teacher at a school with a newly renovated gym introduces a reading fluency routine in January. In May she measures students at 110 words per minute and reports the routine worked. She uses a popular fluency passage set. What is the main problem?

- A. Without a January baseline, she cannot show that fluency changed.
- B. There is no problem; a May score proves improvement.
- C. The passage set is popular, so it must be invalid.
- D. 110 words per minute is too low to count as success.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher tests a new vocabulary routine from September to May. Scores rise. During the same months, the school adopted a new reading curriculum and students grew a year older. The teacher's classroom has a new rug. What should she acknowledge?

- A. The new curriculum (history) and normal growth (maturation) may explain some of the gain.
- B. Nothing can ever be learned from classroom research.
- C. The gain proves the routine caused improvement.
- D. The new rug may have raised scores.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A teacher wants to rule out that her students' gains came from simply growing older during the year. Which design choice helps most?

- A. Measure only at the end of the year.
- B. Survey students about whether they grew.
- C. Use a longer study period.
- D. Compare with a similar class that did not receive the routine over the same months.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher who presents often at local conferences concludes from one end-of-unit survey that a new discussion protocol increased student engagement. The survey was completed on new tablets. How could she strengthen the conclusion?

- A. Give the same survey a second time.
- B. Survey results are always enough by themselves.
- C. Present the finding at a larger conference.
- D. Triangulate with observation tallies of participation and samples of student work.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher plans to publish a study using her students' writing samples and interview quotes. Her principal is enthusiastic, and the district just launched a research newsletter. She tells students that those who take part will earn extra credit. What is the ethical problem?

- A. No problem, since the principal approves.
- B. Tying participation to extra credit pressures students; she also needs required approvals, parental consent and student assent, with confidentiality.
- C. The only issue is that she should use real names to give credit.
- D. Classroom research never needs consent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A parent declines to let her child take part in a teacher's action research study. What should happen?

- A. The whole study must be cancelled.
- B. Use the child's data anyway without the name.
- C. The child continues normal instruction without penalty, and the child's data are not used in the study.
- D. The child is moved to another class.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A fifth-grade teacher ~~at a school with a new outdoor classroom~~ notices many students struggle with multi-step word problems. She wants to try a strategy she heard about at lunch. ~~She has taught for nine years.~~ According to the action research cycle, what should she do before trying the strategy?

Target word: step before acting **Decides:** Define the problem clearly and review the literature before planning an intervention.

A	Inverted operation	Experience is not a step in the cycle; inquiry can start now.
B	Dropped element	Skips problem definition and literature review.
C	Correct	Problem definition and literature review come before action.
D	Camouflage	Answers the setting detail, not the process.

2. Answer C Concept 2

A high school teacher ~~who also coaches track~~ drafts action research questions. ~~The school recently won a regional award.~~ Which question is most researchable in her own setting?

Target word: best question **Decides:** Focused, within her control, answerable with her data.

A	Camouflage	Answers the award detail; not an improvement question.
B	Absolute language	Far too broad for one educator to study.
C	Correct	Specific change, group, outcome and time frame.
D	Adjacent category	Outside her control and framed from a deficit view.

3. Answer B Concept 3

For her literature review, a teacher ~~who reads widely~~ finds four sources on vocabulary instruction. One is on a popular teaching website with over a million followers. Which source provides the strongest research evidence?

Target word: strongest source **Decides:** Peer-reviewed empirical research.

A	Camouflage	Popularity is not peer review.
B	Correct	Empirical, peer-reviewed research is the strongest evidence.
C	Confusion set	Experience-based opinion, not research.
D	Adjacent category	Marketing material, not independent research.

4. Answer A Concept 3 • transfer

A teacher has cited only teacher blogs and a vendor white paper. What should she add to strengthen her literature review?

Target word: what to add **Decides:** Peer-reviewed empirical studies.

A	Correct	They provide research evidence the other sources lack.
B	Absolute language	Sources differ in rigor.
C	Inverted operation	Adds another non-independent source.
D	Adjacent category	More of the same non-research source.

5. Answer D Concept 4

A middle school counselor ~~with a new office~~ asks: "How do students who recently moved to our school describe their experience of making friends?" ~~The school has 850 students.~~ Which method fits best?

Target word: method **Decides:** A "how do students describe" question calls for qualitative data.

A	Wrong stem number	Uses the enrollment figure; not about the question.
B	Adjacent category	Scores do not describe friendship experiences.
C	Absolute language	Not feasible or ethical, and it does not fit the question.
D	Correct	Captures experiences in students' own words.

6. Answer B Concept 4 • transfer

A teacher asks: "By how many points did students' quiz averages change after I added retrieval practice?" Which method fits?

Target word: method **Decides:** "By how many points" calls for quantitative data.

A	Adjacent category	Reviews others' findings, not her students' change.
B	Correct	Measures an amount of change.
C	Inverted operation	Qualitative data cannot give a points change.
D	Confusion set	Also qualitative.

7. Answer A Concept 5

A teacher at a school ~~with a newly renovated gym~~ introduces a reading fluency routine in January. In May she measures students at 110 words per minute and reports the routine worked. ~~She uses a popular fluency passage set.~~ What is the main problem?

Target word: design problem **Decides:** No baseline was collected before the routine.

A	Correct	Change needs a starting point.
B	Absolute language	One point in time cannot show change.
C	Camouflage	Popularity does not decide validity.
D	Wrong stem number	The number is not the flaw; the missing comparison is.

8. Answer A Concept 6

A teacher tests a new vocabulary routine from September to May. Scores rise. During the same months, the school adopted a new reading curriculum and students grew a year older. ~~The teacher's classroom has a new rug.~~ What should she acknowledge?

Target word: limitation **Decides:** Other events and maturation could explain the gain.

A	Correct	Alternative explanations limit how much credit the routine gets.
B	Absolute language	Limits call for caution, not dismissal.
C	Inverted operation	Ignores competing explanations.
D	Camouflage	Not a plausible or relevant explanation.

9. Answer D Concept 6 • transfer

A teacher wants to rule out that her students' gains came from simply growing older during the year. Which design choice helps most?

Target word: design that controls maturation **Decides:** A comparison group of similar students over the same period.

A	Dropped element	Removes the baseline as well.
B	Adjacent category	Perception data do not control the threat.
C	Inverted operation	More time allows more maturation.
D	Correct	Both groups mature; a difference points to the routine.

10. Answer D Concept 7

A teacher ~~who presents often at local conferences~~ concludes from one end-of-unit survey that a new discussion protocol increased student engagement. ~~The survey was completed on new tablets.~~ How could she strengthen the conclusion?

Target word: strengthening credibility **Decides:** Add other data sources on engagement.

A	Adjacent category	Repeats one source rather than adding another.
B	Absolute language	One self-report source is limited.
C	Camouflage	Audience size does not strengthen evidence.
D	Correct	Multiple sources that agree strengthen findings.

11. Answer B Concept 8

A teacher plans to publish a study using her students' writing samples and interview quotes. ~~Her principal is enthusiastic, and the district just launched a research newsletter.~~ She tells students that those who take part will earn extra credit. What is the ethical problem?

Target word: ethical issue **Decides:** Participation is tied to grades, making it coercive; consent and approvals are also required.

A	Camouflage	Enthusiasm is not the formal approval and consent required.
B	Correct	Participation must be voluntary and free of penalty or reward in grades.
C	Inverted operation	Confidentiality requires protecting identities.
D	Absolute language	Publishing student data requires consent and approval.

12. Answer C Concept 8 • transfer

A parent declines to let her child take part in a teacher's action research study. What should happen?

Target word: response to a declined consent **Decides:** No penalty; the child's data are excluded.

A	Absolute language	The study can continue with consenting participants.
B	Dropped element	Leaves out consent; anonymity does not replace it.
C	Correct	Declining must carry no consequence.
D	Inverted operation	Penalizes the family's choice.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	2	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The action research cycle	Action research moves through identifying a problem, reviewing literature, planning an intervention, acting and collecting data, analyzing, and reflecting to plan the next cycle; it is cyclical, not one-and-done.	Jumping to an intervention before defining the problem or reviewing literature, or stopping after one round.	A question about what step comes next.
2	A researchable question	A good action research question is focused, names what will change and for whom, and can be answered with data the educator can collect in their own setting.	Asking a question too broad to study or about something outside the educator's control.	A draft research question to judge.
3	Scholarly vs. non-scholarly sources	Peer-reviewed empirical studies are the strongest literature for informing practice; blogs, vendor materials and opinion pieces can inform but are not research evidence.	Treating a popular article or vendor claim as research.	A list of sources for a literature review.
4	Match the method to the question	Quantitative methods answer how much or how many (scores, counts); qualitative methods answer how or why from people's experiences (interviews, observations); mixed methods combine both.	Using only test scores to answer a question about students' experiences, or the reverse.	The wording of the research question: "how much" vs. "how" or "why."
5	Baseline data	Collect data before the intervention so change can be measured against a starting point.	Measuring only after the intervention and calling the result improvement.	A plan that collects data only at the end.
6	Alternative explanations	Change during a study may come from other events (history), normal development (maturation), or practice with the test; strong designs and honest discussion account for these.	Crediting the intervention for all change seen.	Something else changed during the study period.
7	Triangulation	Using several data sources or methods to answer the same question strengthens the credibility of findings.	Relying on a single survey or single score.	One data source supports a big conclusion.
8	Research ethics in schools	Educator-researchers get required approvals, obtain informed consent (and student assent), protect confidentiality, and never penalize students or families for declining to participate in the research.	Assuming classroom research needs no permission, or tying participation to grades.	A plan to collect or publish data about students.

Course scope (public description): Educational Inquiry focuses on practical problem solving. This course teaches candidates to use scholarly literature to inform their own practice. It also teaches candidates to engage in their own action research processes, which empowers educators to recognize opportunities for improvement and to systematically implement and evaluate changes. This course prepares candidates to conduct research for the capstone. Prerequisites for this course: Data Literacy and Evidence-Based Practices.